



NUI Galway
OÉ Gaillimh

Semester 2 Examinations 2019/2020

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This Examination, MP307, is an Open Book examination.

Exam Code(s)	3BMS2, 3BPT1, 3BS9, 3BPT1, 3FM2, 4BME1, 4BMS2, 4BS2
Exam(s)	3 rd Arts, 3 rd & 4 th Science
Module Code(s)	MP307
Module(s)	Modelling II
Paper No.	1
Repeat Paper	No
External Examiner(s)	Professor Y. Fu
Internal Examiner(s)	Dr P. Piironen, Dr M. Meere

Instructions: Marks are awarded for question 1 in Section A **and** the TWO best questions in Section B.
All questions carry the same weight.

Duration	2 hours
No. of Pages	6
Discipline(s)	Applied Mathematics
Course Co-ordinator(s)	Dr P. Piironen

Requirements:

Release in Exam Venue	Yes
Statistical/ Log Tables	Yes
Other Materials	Calculator

Section A - Answer all parts (a)-(d). (Each part is worth 5 Marks.)

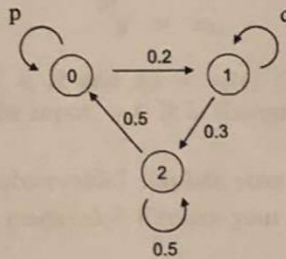
~~1.~~ (a) Let

$$f(t) = \begin{cases} \frac{2t+1}{4}, & 1 \leq t \leq 2 \\ 0, & 0 \leq t < 1, t > 2 \end{cases}$$

be the probability density function for the waiting time in a queue.

- ~~(i)~~ Verify that $f(t)$ is a probability density function.
- ~~(ii)~~ What is the average waiting time?
- ~~(iii)~~ What is the variance and the standard deviation of the waiting time?

~~(b)~~ Consider the following Markov Process with transition probabilities shown.



- ~~(i)~~ Find the transition matrix P and the probabilities p and q .
- ~~(ii)~~ Is the Markov process ergodic? Explain your answer.
- ~~(iii)~~ Does the Markov process have unique equilibrium probabilities? Explain your answer. If the Markov process has unique equilibrium probabilities calculate them.

PTO

Section B - Answer any two questions.

2. The size of queue is given by a nearest-neighbour Markov model. It has a Poisson arrival pattern with parameter α_k and a Poisson servicing pattern with parameter β_k , where k is the queue size.

(a) Let the maximum size of the queue be r .

(i) Explain why $\alpha_r = \beta_0 = 0$.

(ii) Assuming that the Markov Process is ergodic, show that the equilibrium probabilities are given by

$$\pi_k = \rho_k \rho_{k-1} \dots \rho_1 \pi_0$$

for $0 < k \leq r$ where $\rho_k = \alpha_{k-1} / \beta_k$.

8 Marks

(b) Let the queue be infinite with ample servers such that

$$\alpha_k = \alpha, \quad \beta_k = k\beta,$$

where $\alpha > 0$ and $\beta > 0$.

Show that the equilibrium probabilities π_k , $k = 0, 1, 2, \dots$, are given by the Poisson distribution

$$\pi_k = \frac{\rho^k e^{-\rho}}{k!}, \quad \rho = \frac{\alpha}{\beta}.$$

6 Marks

(c) What is the probability that the queue in (b) is longer than 3 if

$$\alpha = 300 \text{ hour}^{-1} \quad \text{and} \quad \beta = 4 \text{ min}^{-1}?$$

6 Marks

PTO

3. Consider an insect species in which no individuals live beyond 3 days. Divide the insect population into four age groups labelled by $n = 0, 1, 2$ and 3. Assume that only the second ($n = 1$) and third ($n = 2$) groups can reproduce. For each group n , let b_n denote the birth rate and d_n the death rate.

- (a) Find a matrix A such that $P(t+1) = AP(t)$ with

$$P(t) = \begin{pmatrix} P_0(t) \\ P_1(t) \\ P_2(t) \\ P_3(t) \end{pmatrix},$$

where $P_n(t)$ denotes the population size of age group n at day t .

4 Marks

- (b) Show that the eigenvalues of A satisfy the equation

$$\lambda^4 - b_1(1-d_0)\lambda^2 - b_2(1-d_0)(1-d_1)\lambda = 0.$$

4 Marks

- (c) If $d_0 = 0.2$, $d_1 = 0.1$, $d_2 = 0.1$, and $b_1 = 0.9$, $b_2 = 0.4$, show that there is one eigenvalue greater than 1. What is the significance of this?

7 Marks

- (d) Consider the case in (c) over three days, where the initial cohort populations are $P_0 = P_1 = 100$ and $P_2 = P_3 = 50$. How do the results agree with your conclusions in (c)? Discuss briefly.

5 Marks

PTO

4. A two-compartment control system for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u, \quad y = c_2,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, u determines the flow rate of the drug into compartment 1 and y is the measurement of the drug concentration in compartment 2.

- (a) Investigate if the system is observable and/or reachable.

5 Marks

- (b) Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1, k_2, k_r are the control parameters, such that the desired output $y = r = 2$ and the closed loop eigenvalues for the system are $\lambda_1 = -1$ and $\lambda_2 = -1$.

15 Marks

~~(c)~~ Consider a model of two competing species $P_1(t)$ and $P_2(t)$ given by

$$\begin{aligned}\frac{dP_1}{dt} &= 40P_1 - 2P_1^2 - 2P_1P_2, \\ \frac{dP_2}{dt} &= 30P_2 - 3P_1P_2 - 3P_2^2.\end{aligned}$$

~~(i)~~ Give an interpretation of the different terms in the model.

~~(ii)~~ Find all equilibrium points for P_1 and P_2 .

~~(d)~~ Consider the linear control system

$$\begin{aligned}\frac{dx_1}{dt} &= bx_2, \\ \frac{dx_2}{dt} &= x_1 + x_2 + u, \\ y &= x_2,\end{aligned}$$

where $x_1 = x_1(t) \in \mathbb{R}$ and $x_2 = x_2(t) \in \mathbb{R}$ represent the state, $u = u(x_1, x_2) \in \mathbb{R}$ is the input, $y \in \mathbb{R}$ is the output and $b \in \mathbb{R}$ is the control parameter.

~~(i)~~ Is this system observable? Explain your answer.

~~(ii)~~ Is this system reachable? Explain your answer.

PTO